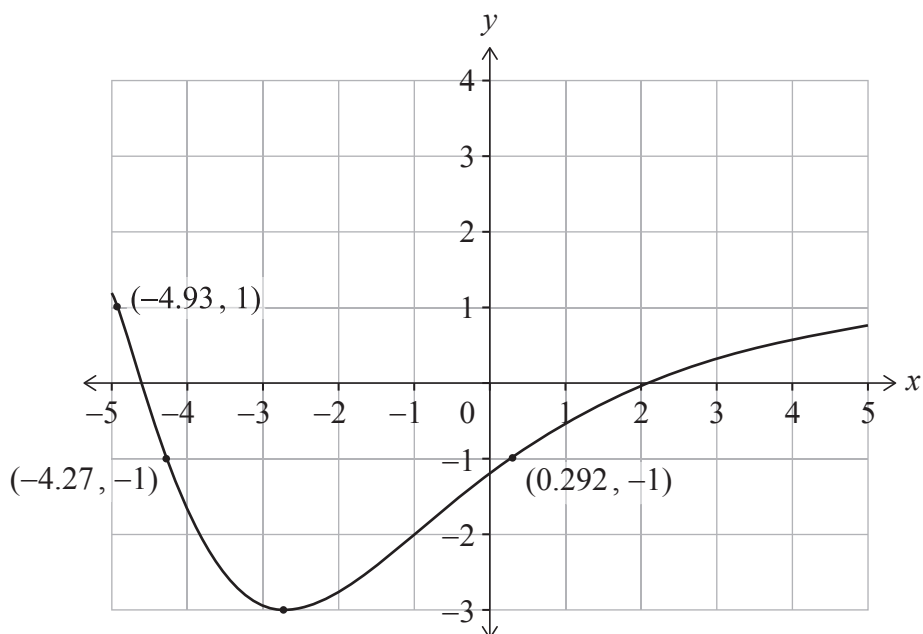


5. [Maximum mark: 8]

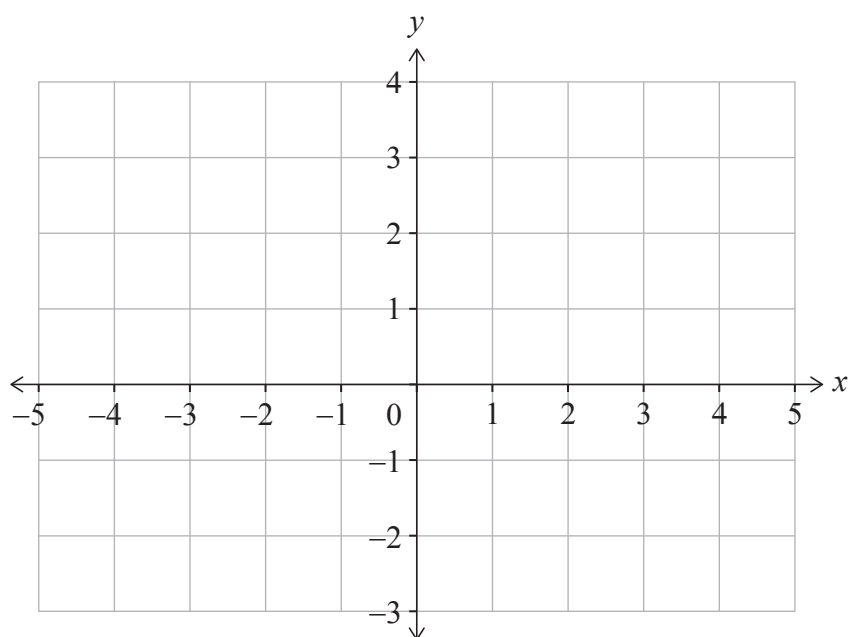
The graph of $y = f(x)$ for $-5 \leq x \leq 5$ is shown in the following diagram.

The curve passes through the points $(-4.93, 1)$, $(-4.27, -1)$ and $(0.292, -1)$.



(a) On the following set of axes,

(i) sketch the graph $y = f(|x|)$ showing the approximate intercepts with the axes;

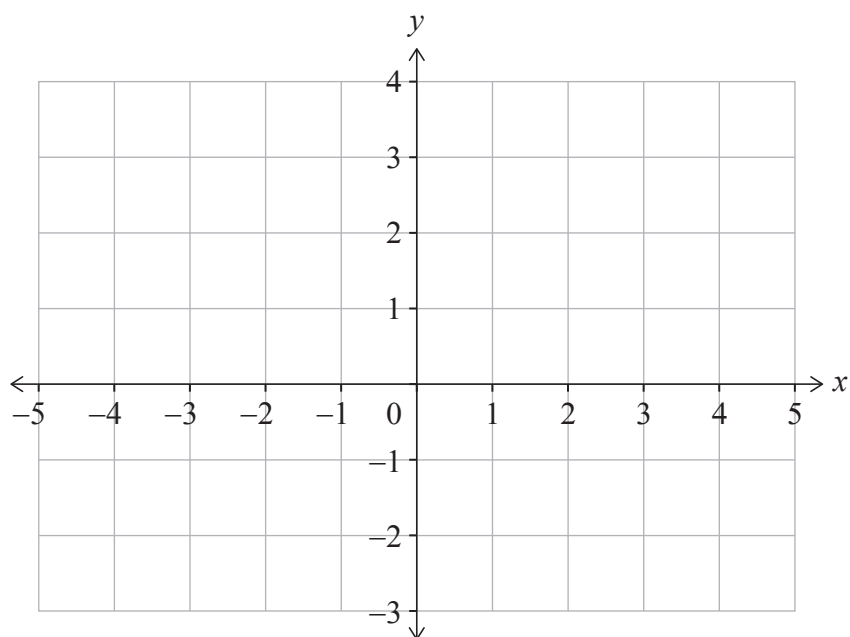


(This question continues on the following page)



(Question 5 continued)

- (ii) sketch the graph $y = |f(x)|$ showing the approximate intercepts with the axes. [4]



- (b) Solve the inequality $|f(x)| \geq (f(x))^2$. [4]

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